THE HYPERBOLA

CONICAL SECTIONS

prepared by:

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Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

40%

WEEKLY

60%

EXAM

PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

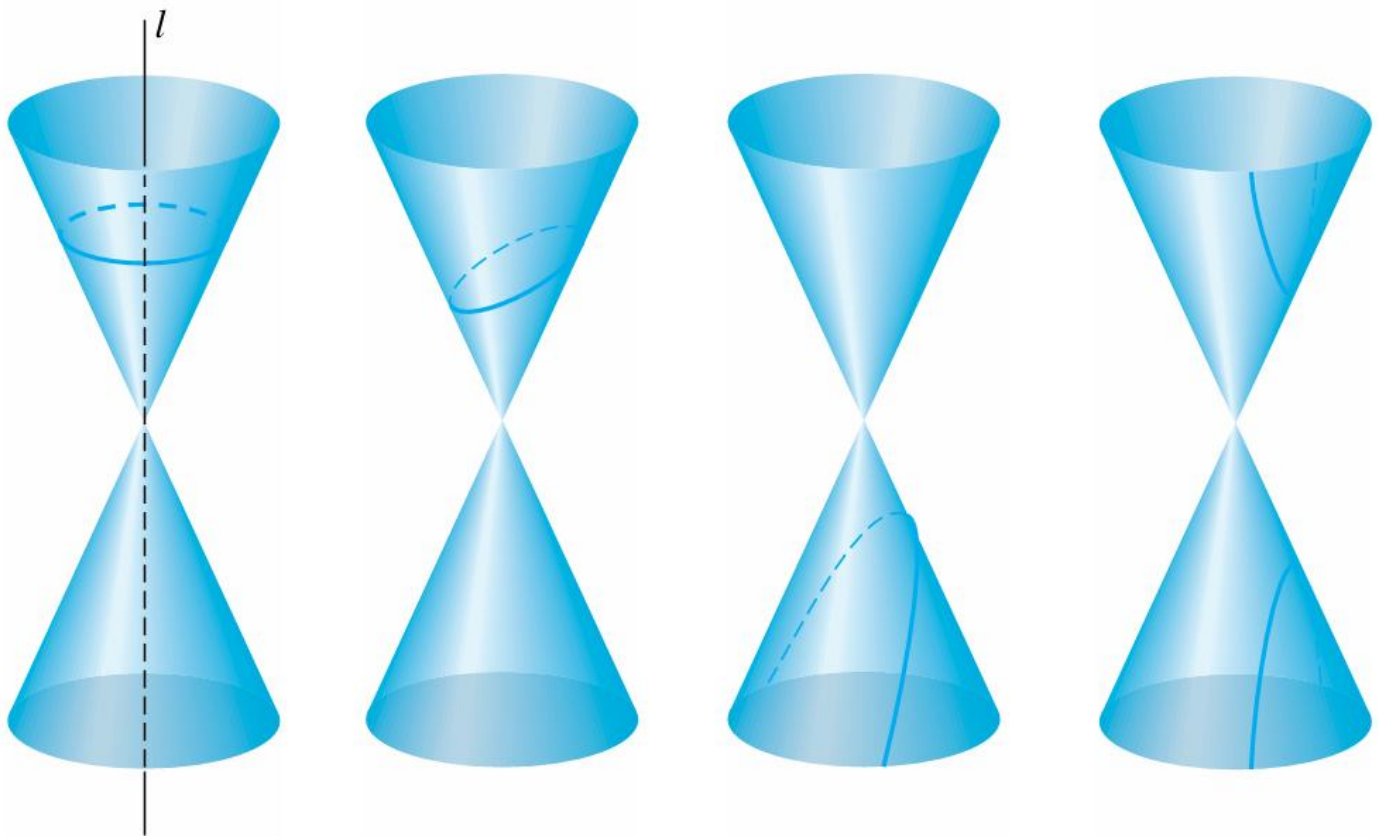
TOPIC OUTLINE

Hyperbola with Center at the Origin

Hyperbola with Center at $C(h, k)$

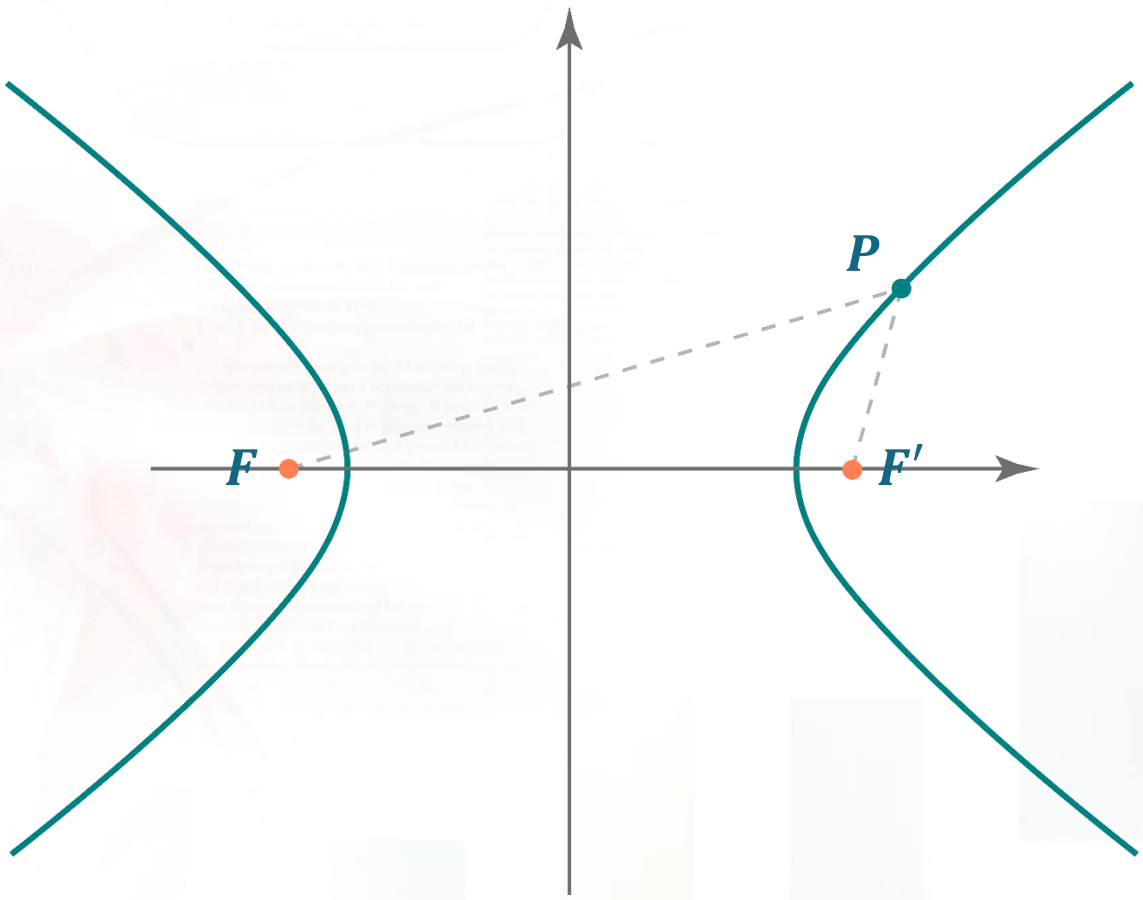
HYPERBOLA WITH CENTER AT THE ORIGIN

CONICAL SECTIONS



Circle Ellipse Parabola **Hyperbola**

A **hyperbola** is the set of all points in a plane, the **difference** of whose distances from two fixed points (the **foci**) in the plane is a positive **constant** ($2a$).



$$d(P, F) - d(P, F') = 2a$$

CENTER AT THE ORIGIN

center

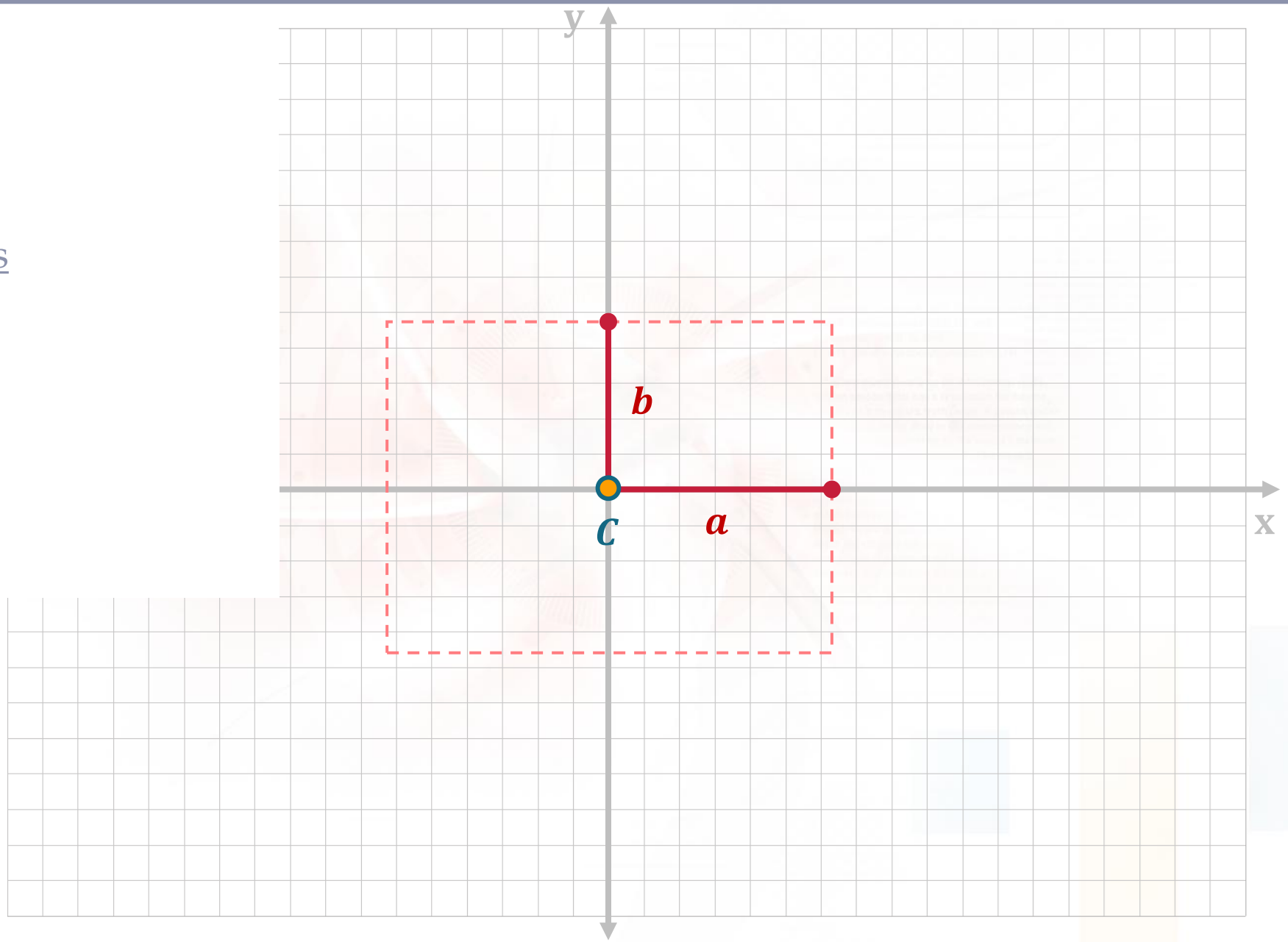
$C(0,0)$

transverse axis

$2a$

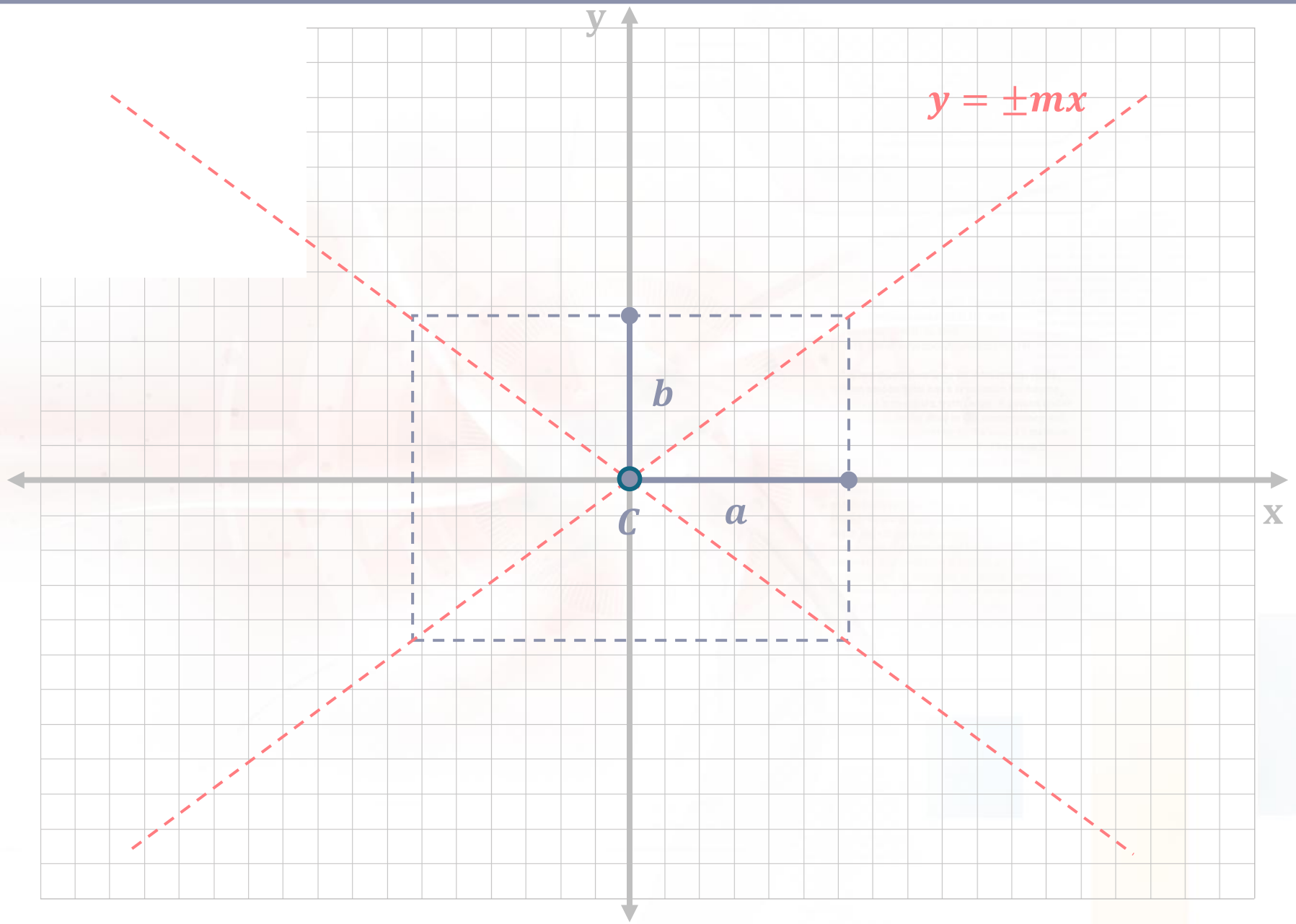
conjugate axis

$2b$



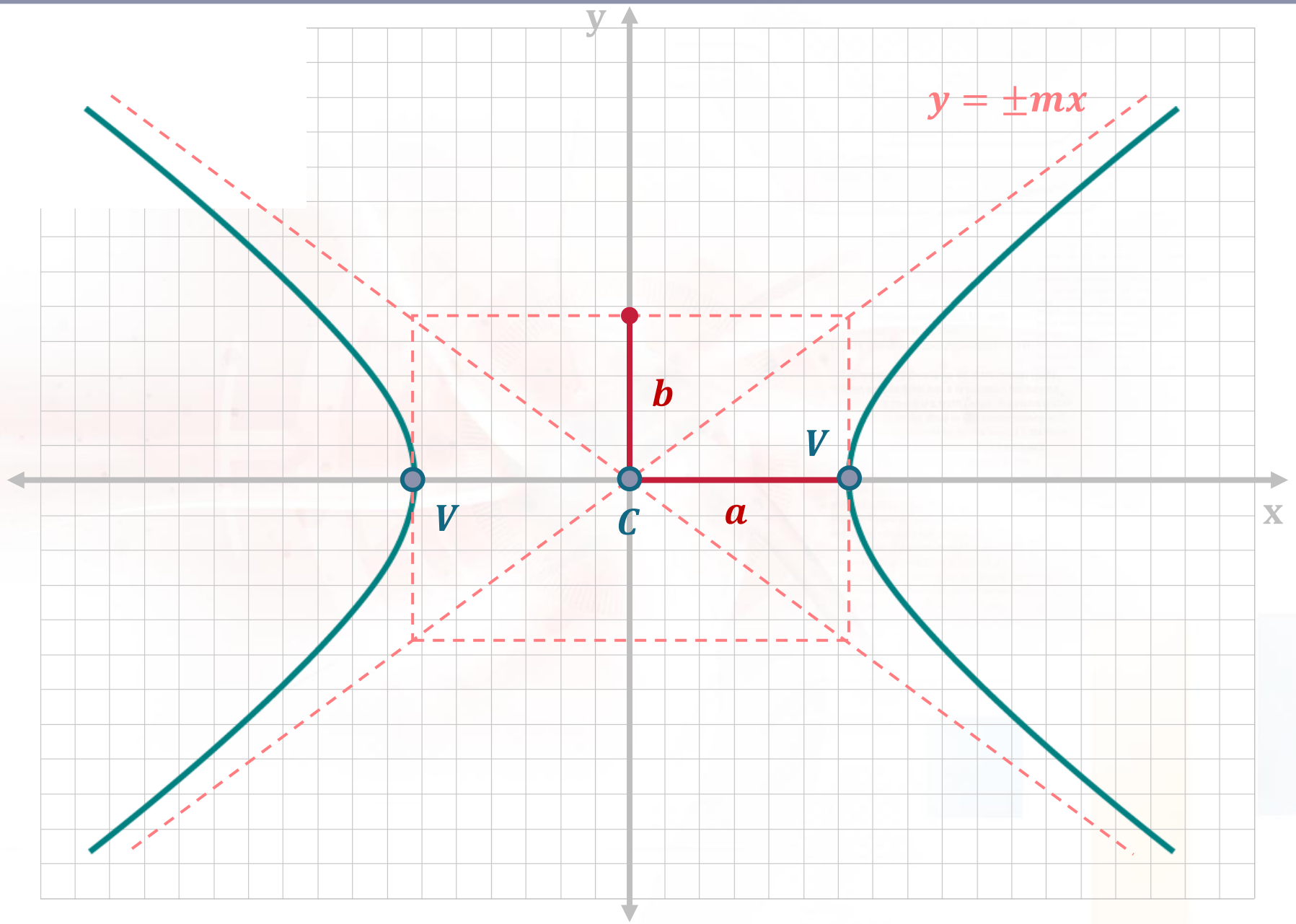
asymptotes

$$y = \pm \frac{b}{a} x$$



vertices

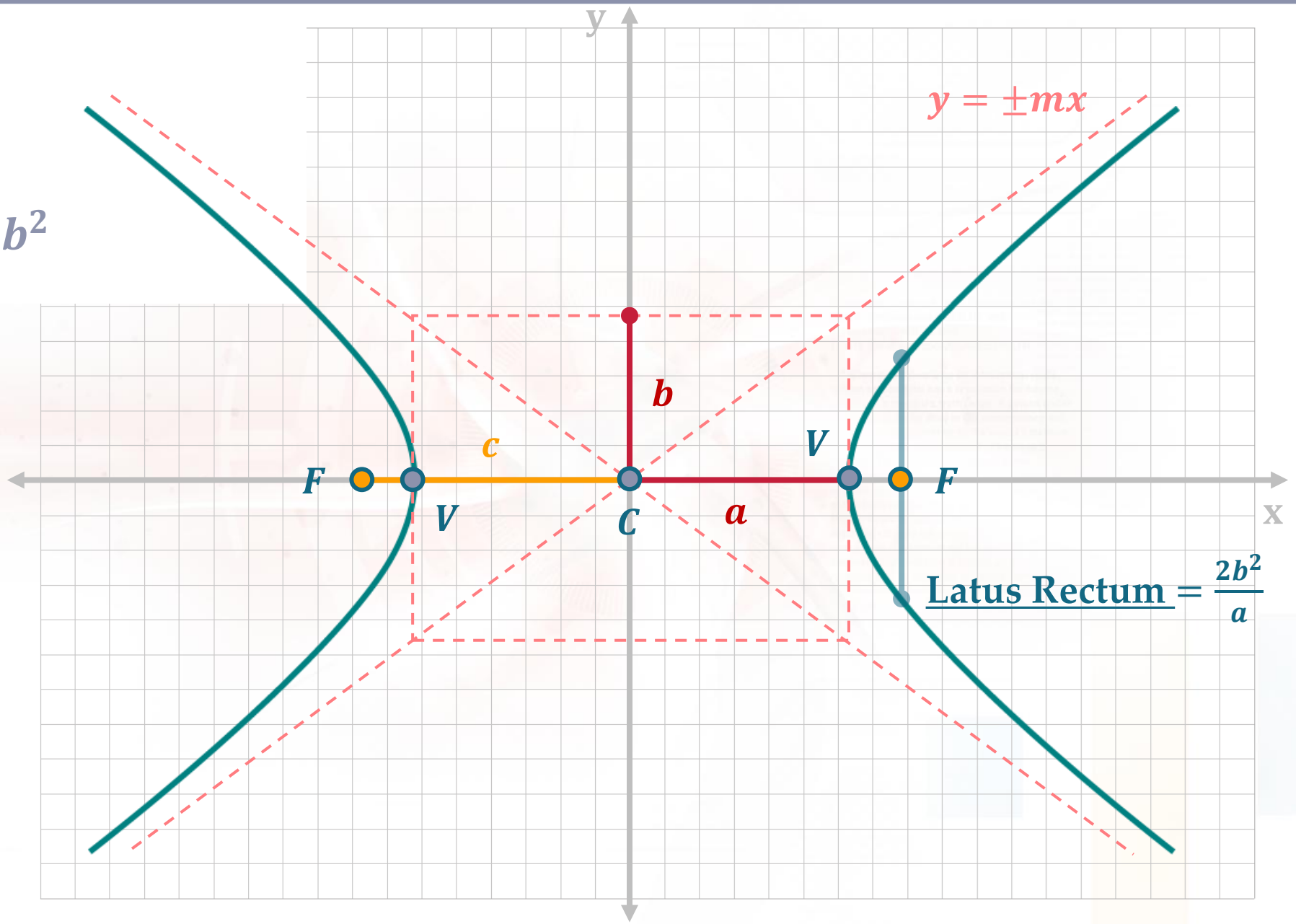
$V(\pm a, 0)$



foci

$F(\pm c, 0)$

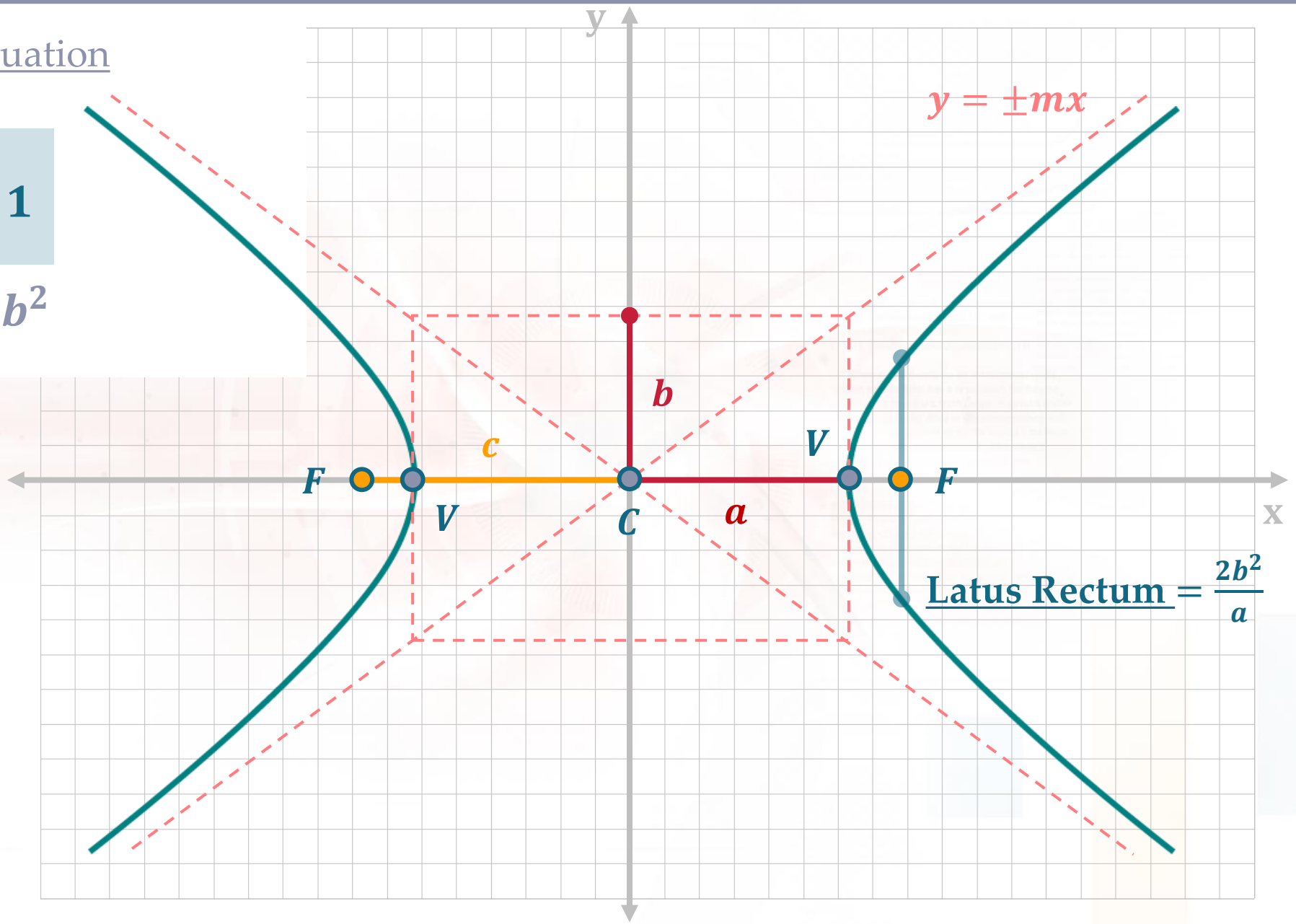
$c^2 = a^2 + b^2$



Standard Equation

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$c^2 = a^2 + b^2$$



TRANSVERSE AXIS is on the $x - axis$

Standard Equation

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$c^2 = a^2 + b^2$$

center, $C(0, 0)$

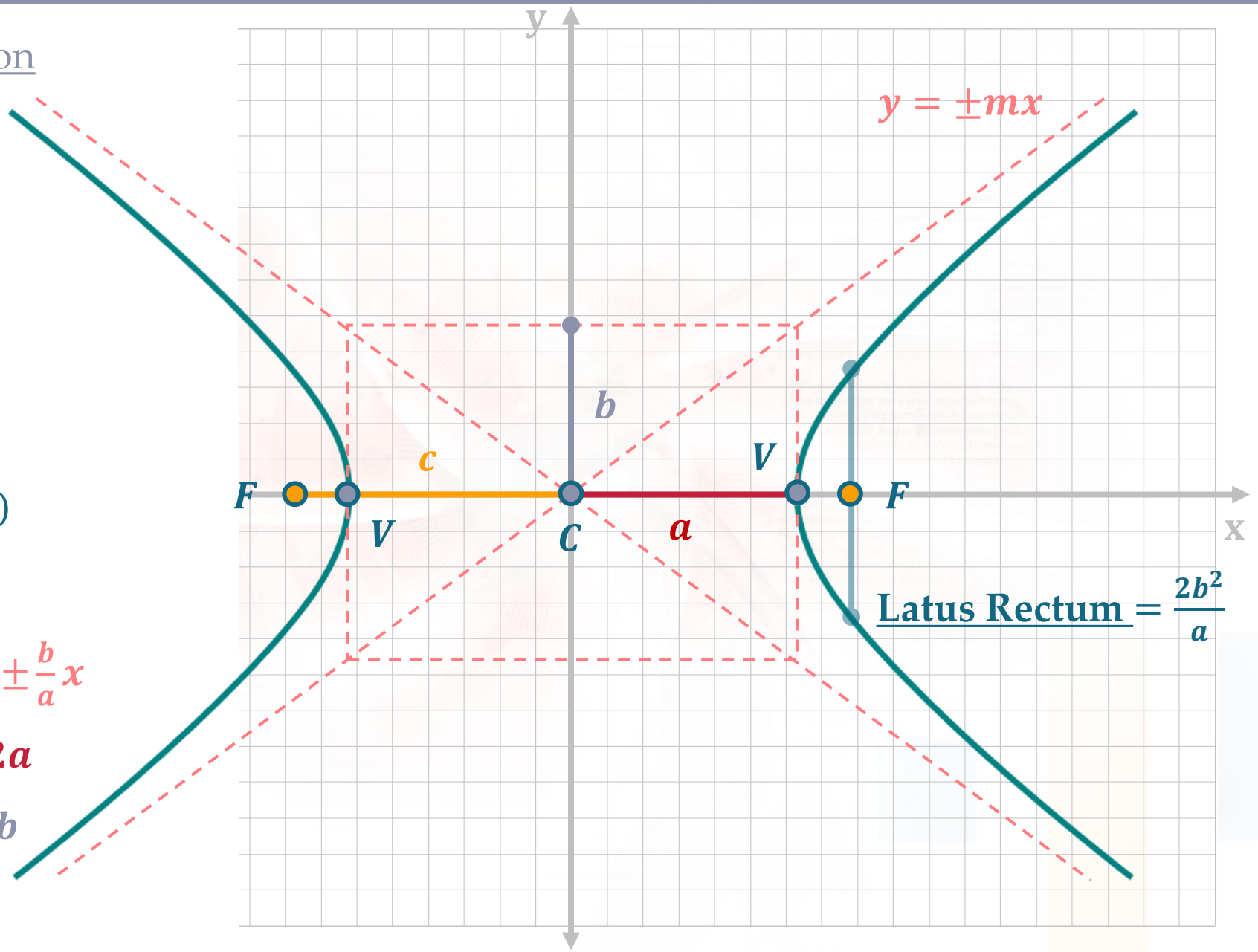
vertices, $V(\pm a, 0)$

foci, $F(\pm c, 0)$

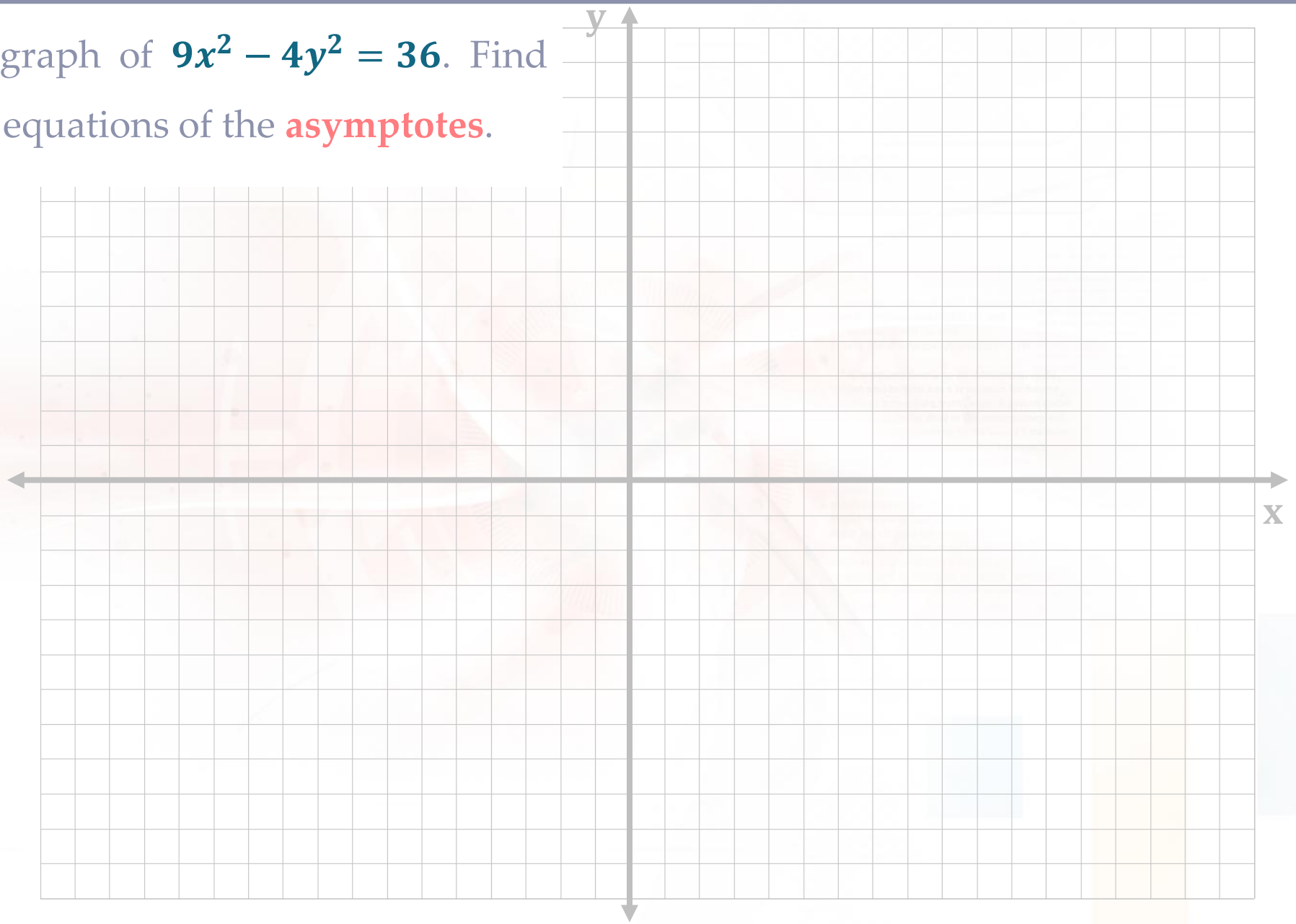
asymptotes, $y = \pm \frac{b}{a} x$

transverse axis, $2a$

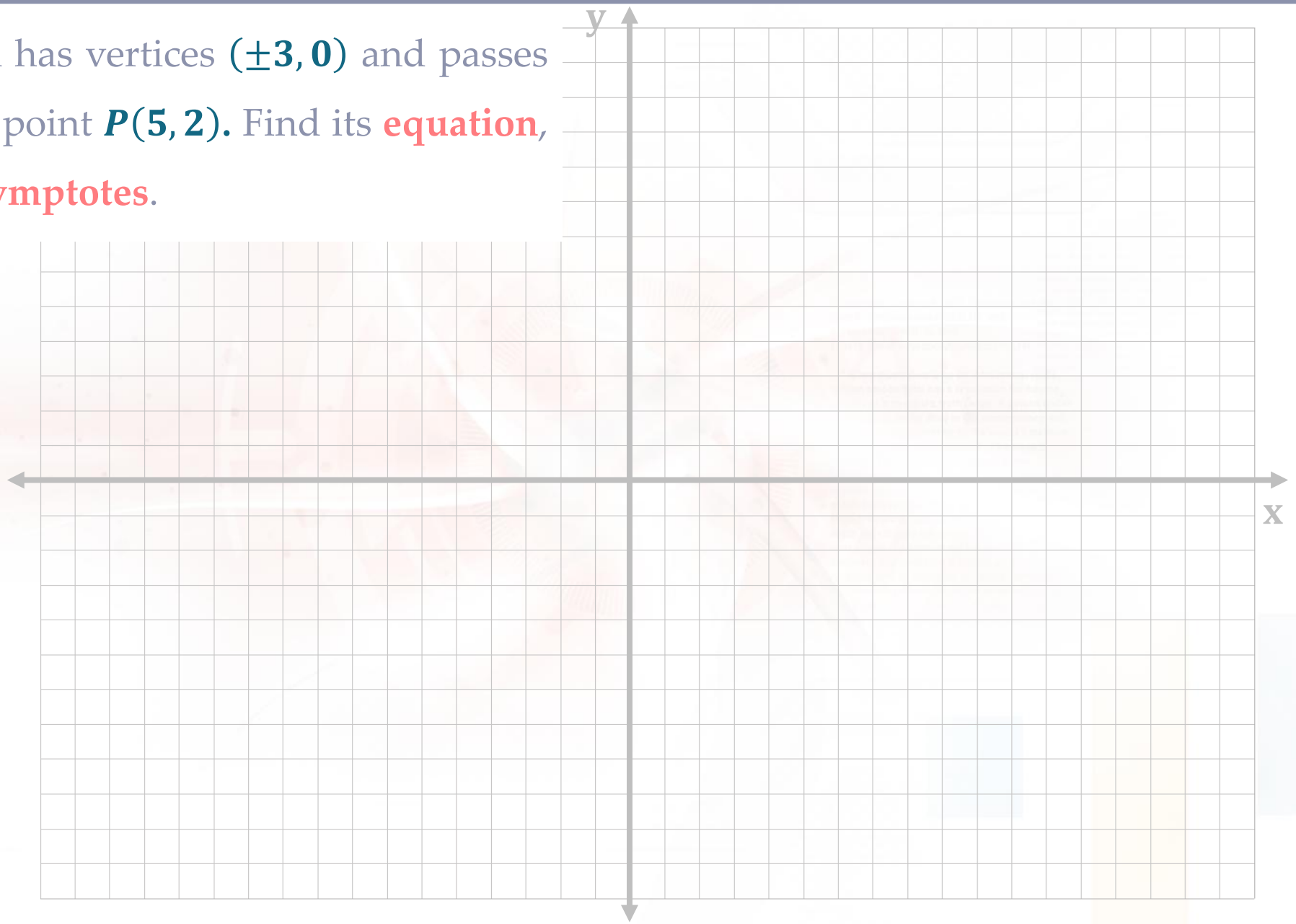
conjugate axis, $2b$



Sketch the graph of $9x^2 - 4y^2 = 36$. Find the **foci** and equations of the **asymptotes**.



A hyperbola has vertices $(\pm 3, 0)$ and passes through the point $P(5, 2)$. Find its **equation**, **foci**, and **asymptotes**.



TRANSVERSE AXIS is on the $y-axis$

Standard Equation

$$\frac{y^2}{a^2}-\frac{x^2}{b^2}=1$$

$$c^2=a^2+b^2$$

center, $C(0,0)$

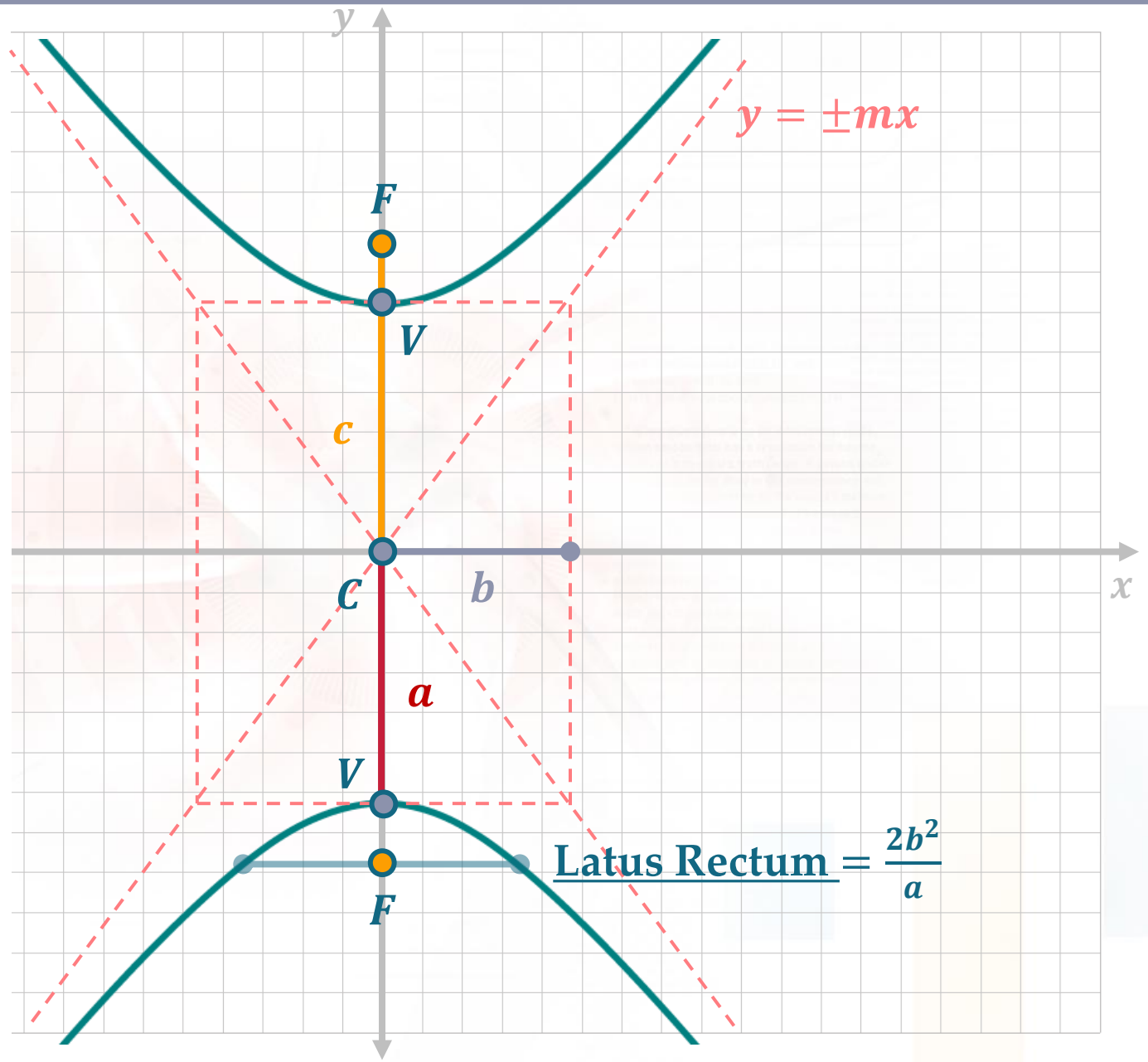
vertices, $V(0,\pm a)$

foci, $F(0,\pm c)$

asymptotes, $y=\pm\frac{a}{b}x$

transverse axis, $2a$

conjugate axis, $2b$



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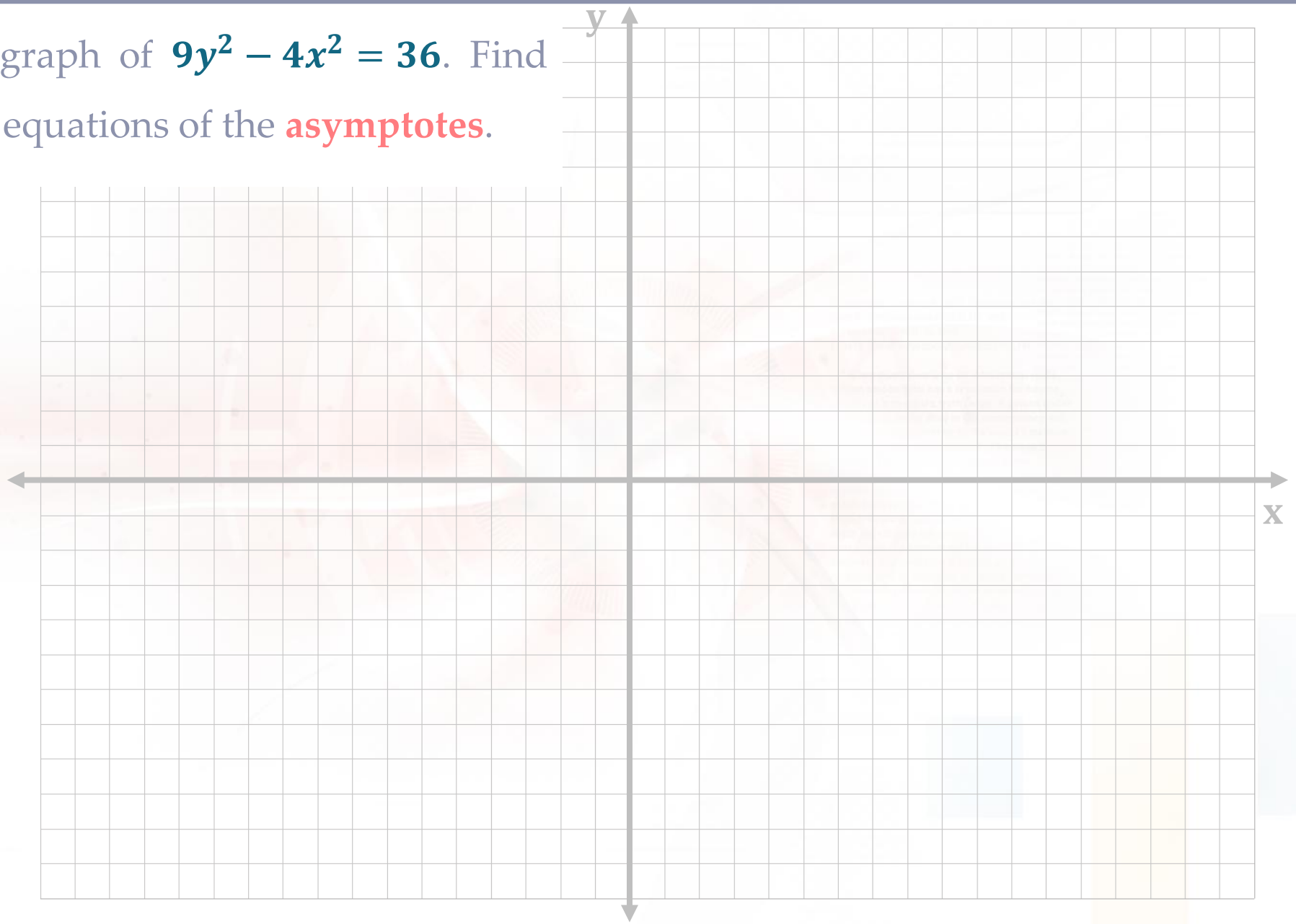
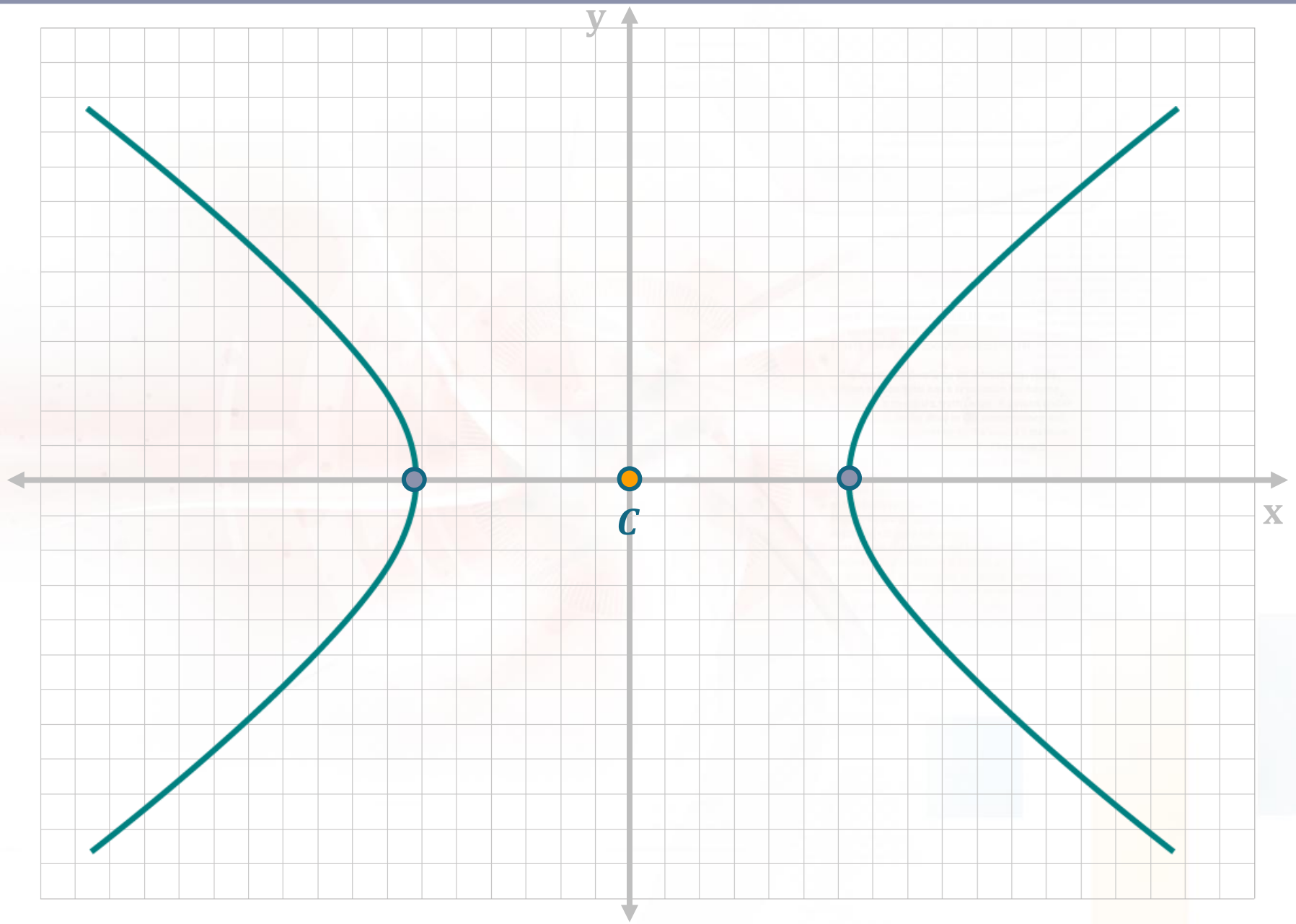


HYPERBOLA WITH CENTER AT $C(h, k)$

CENTER AT THE ORIGIN

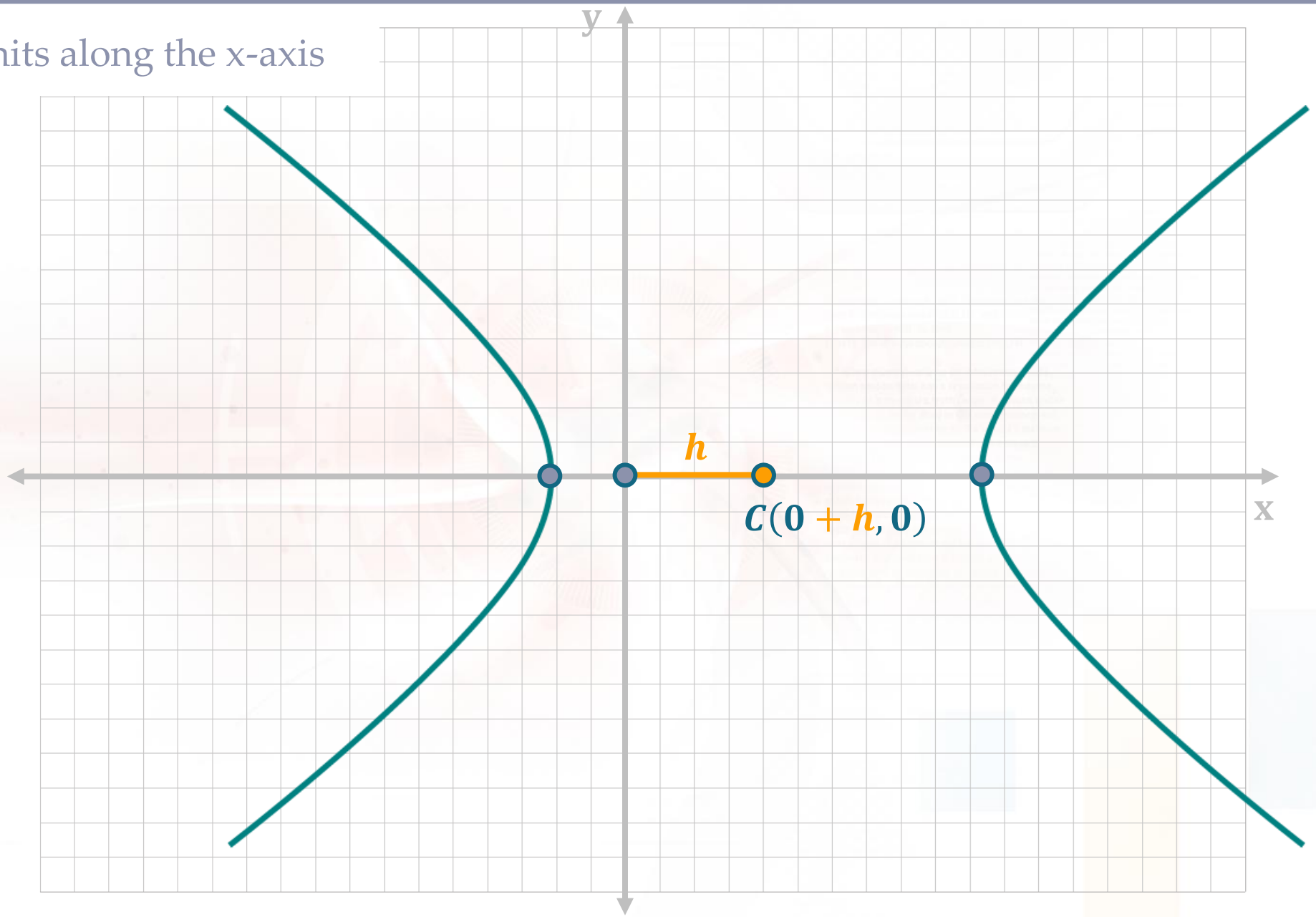
center

$C(0, 0)$



CENTER AT $C(h, 0)$

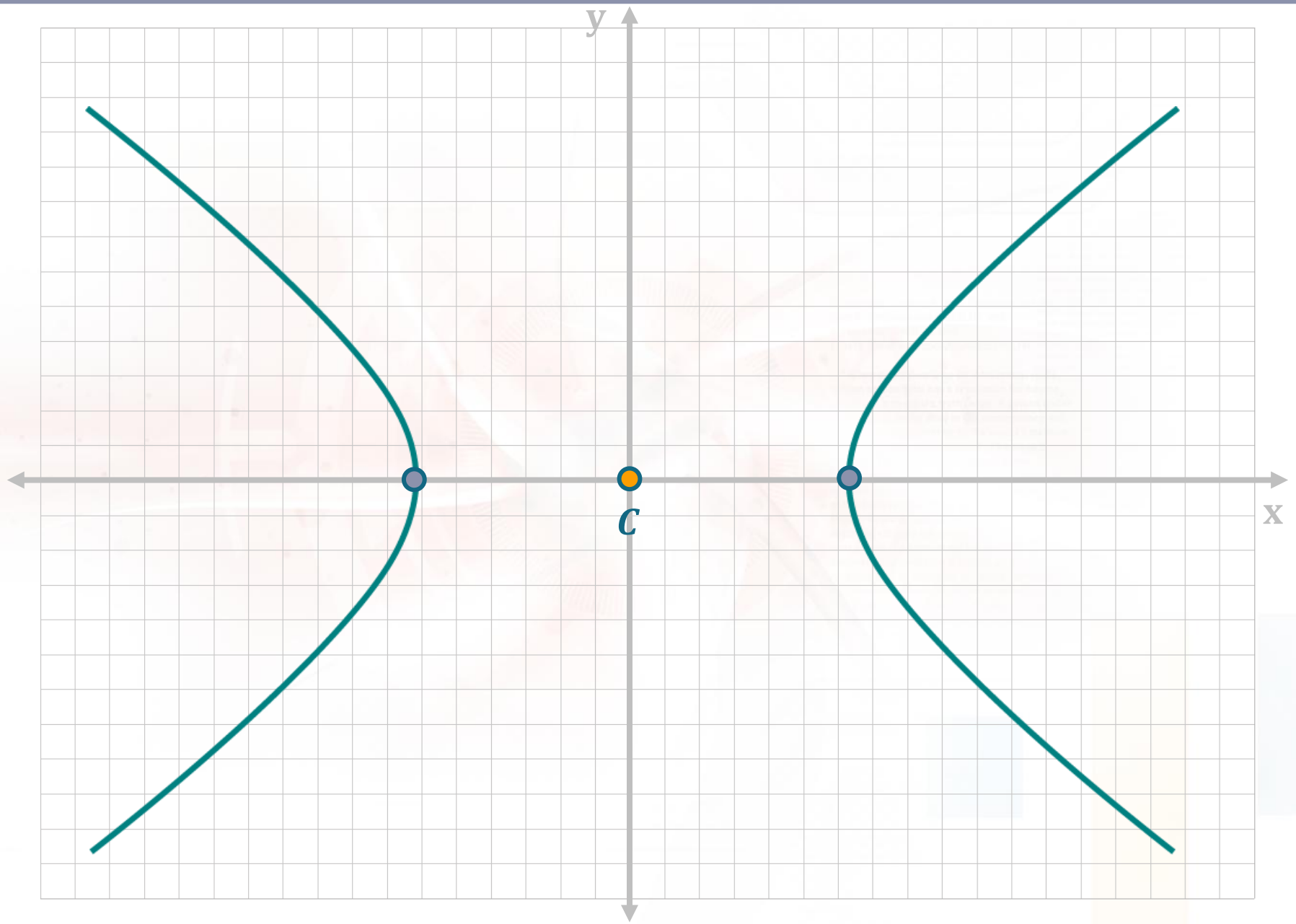
moving h units along the x-axis



CENTER AT THE ORIGIN

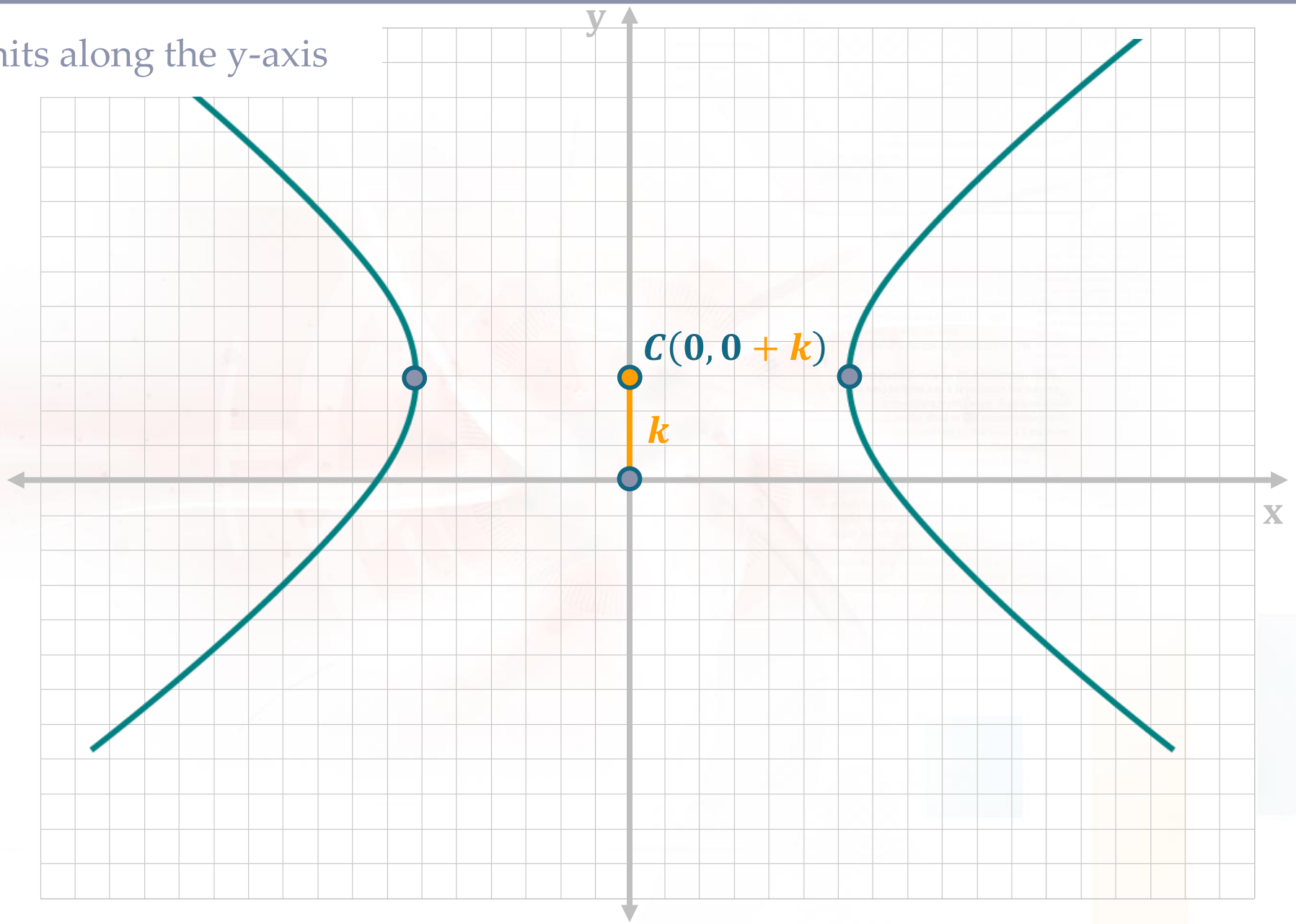
center

$C(0, 0)$



CENTER AT $C(0, k)$

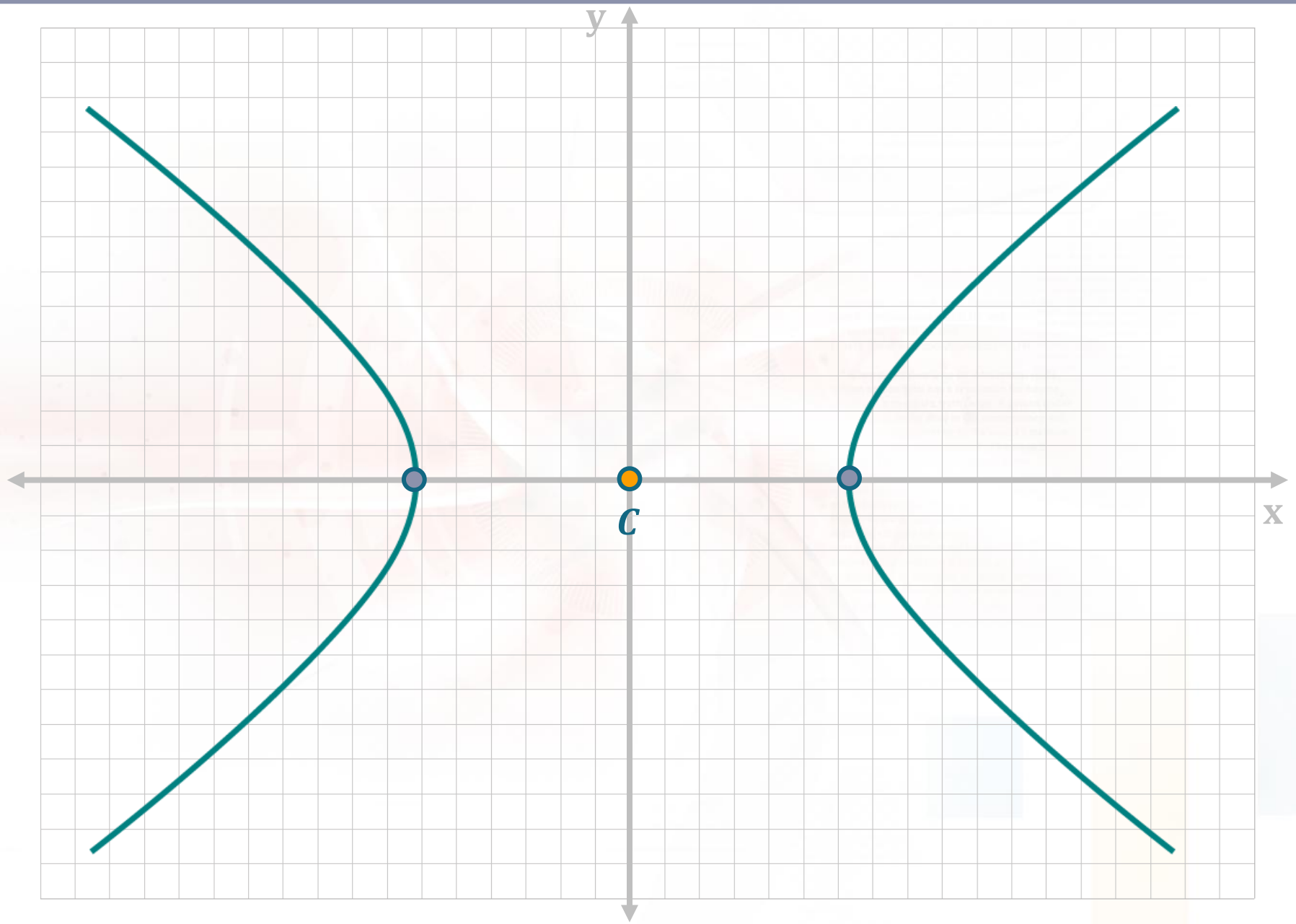
moving k units along the y-axis



CENTER AT THE ORIGIN

center

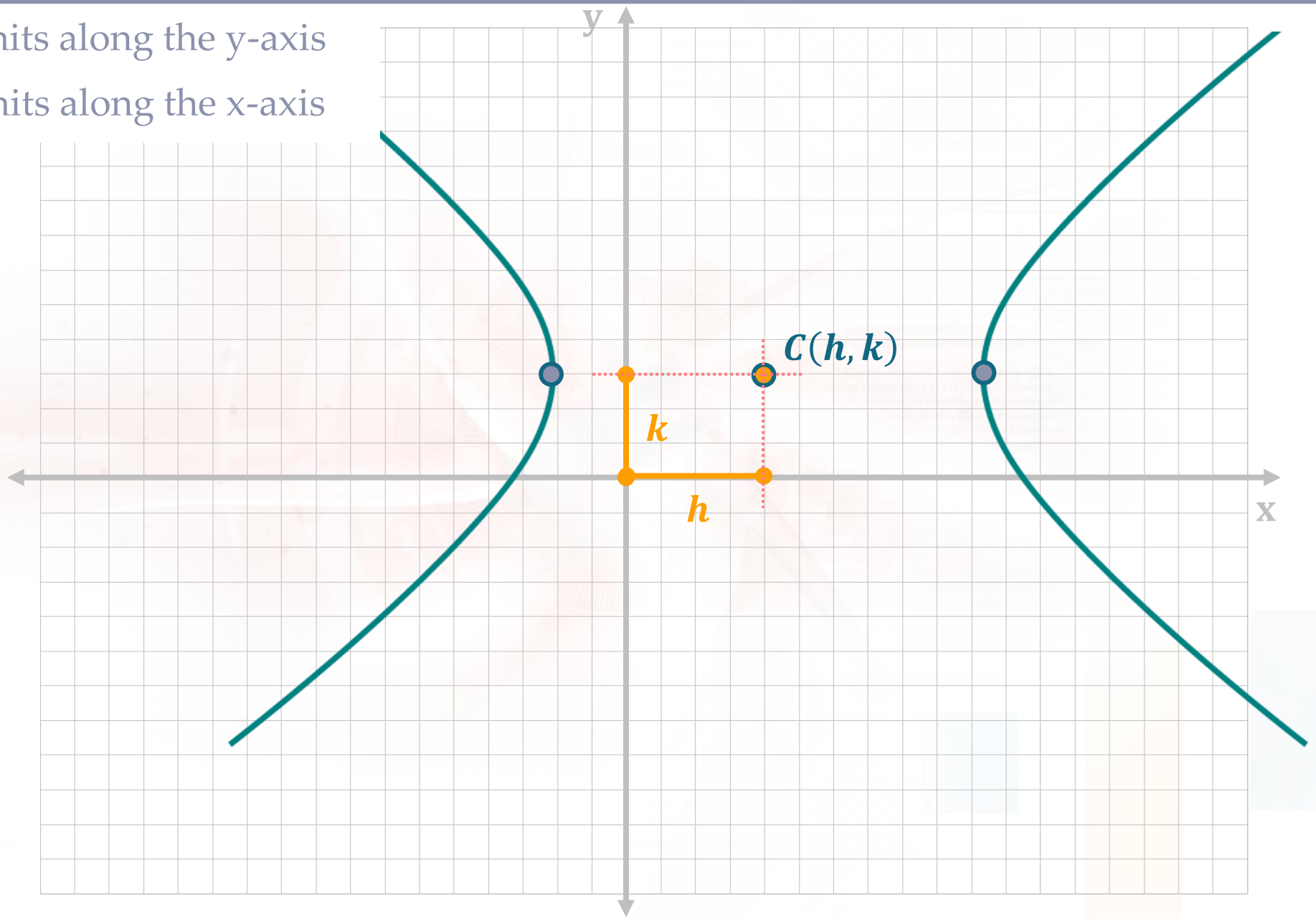
$C(0, 0)$



CENTER AT $C(h, k)$

moving k units along the y-axis

moving h units along the x-axis



TRANSVERSE AXIS is along the $x - axis$

Standard Equation

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

$$c^2 = a^2 + b^2$$

center, $C(h, k)$

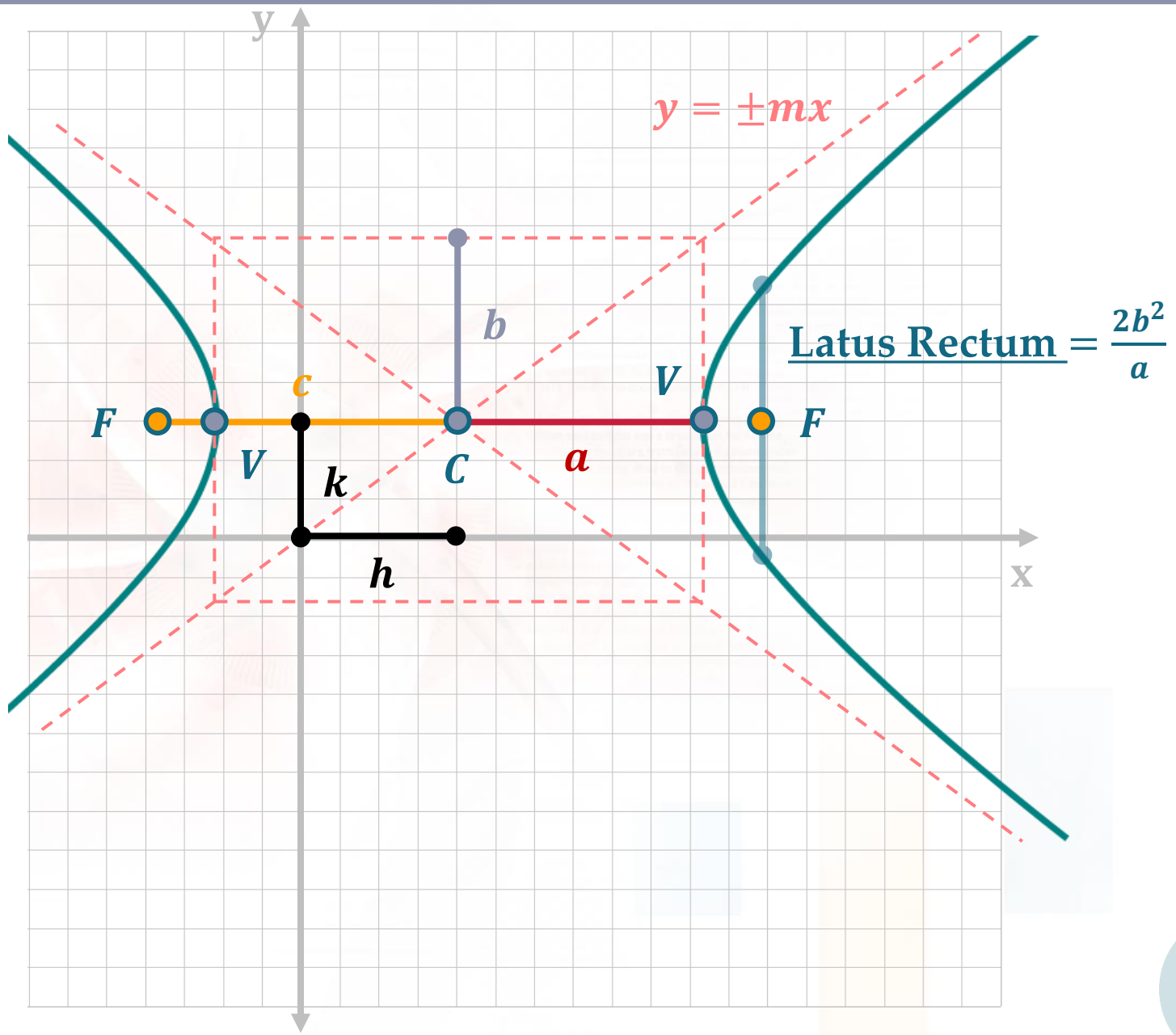
vertices, $V(h \pm a, k)$

foci, $F(h \pm c, k)$

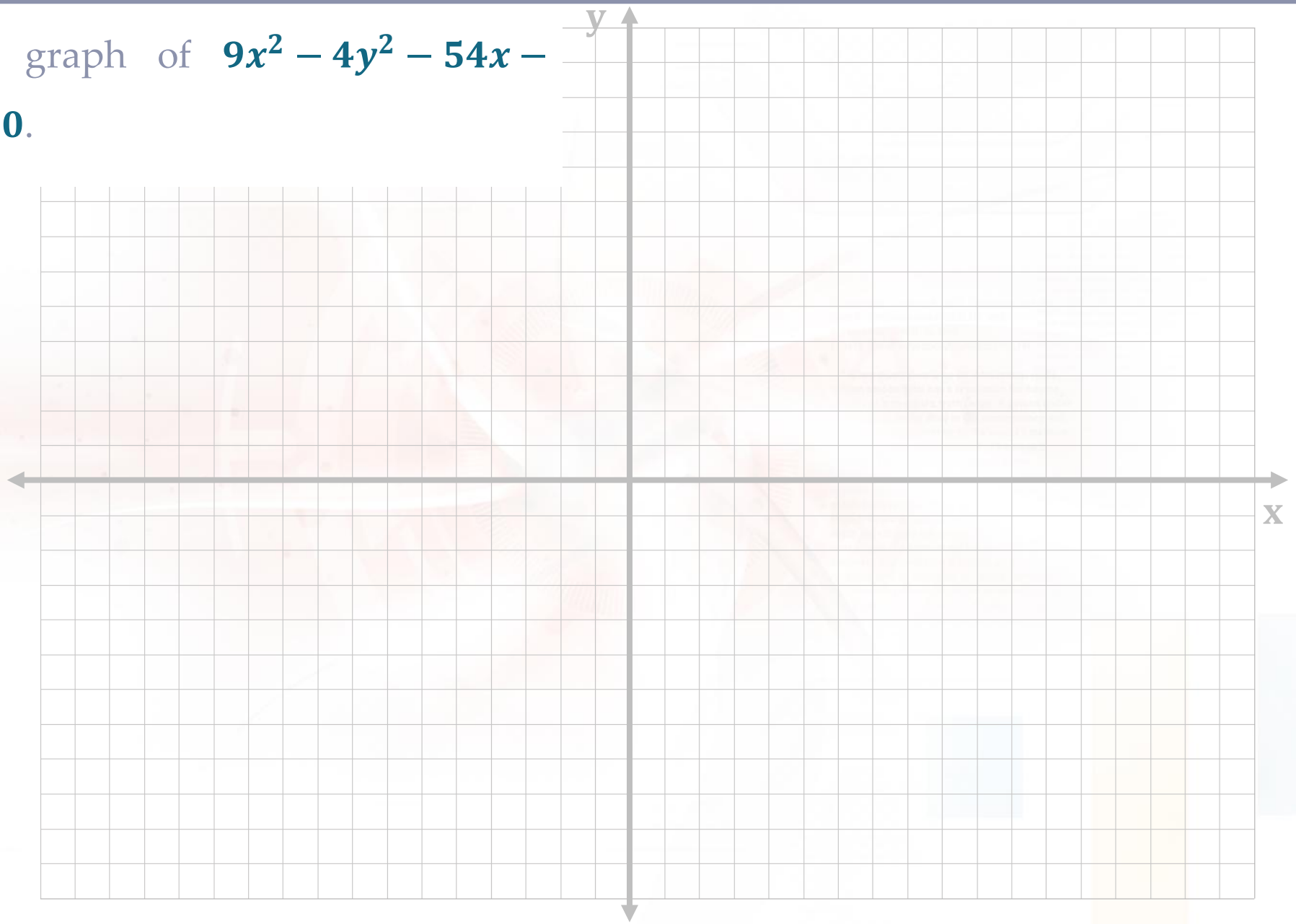
asymptotes, $(y - k) = \pm \frac{b}{a}(x - h)$

transverse axis, $2a$

conjugate axis, $2b$



Sketch the graph of $9x^2 - 4y^2 - 54x - 16y + 29 = 0$.



TRANSVERSE AXIS is along the $y - axis$

Standard Equation

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

$$c^2 = a^2 + b^2$$

center, $C(h, k)$

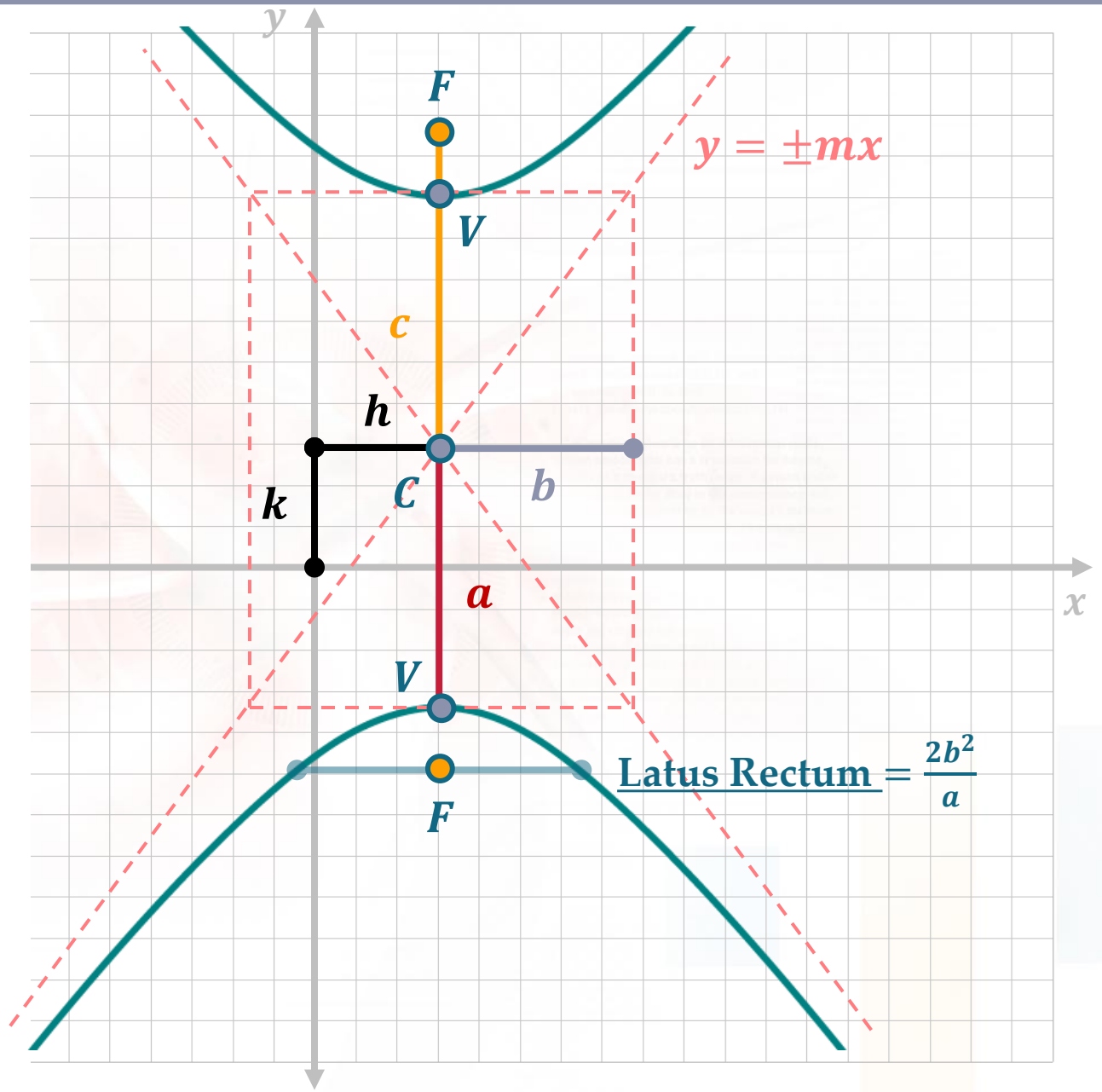
vertices, $V(h, k \pm a)$

foci, $F(h, k \pm c)$

asymptotes, $(y - k) = \pm \frac{a}{b}(x - h)$

transverse axis, $2a$

conjugate axis, $2b$



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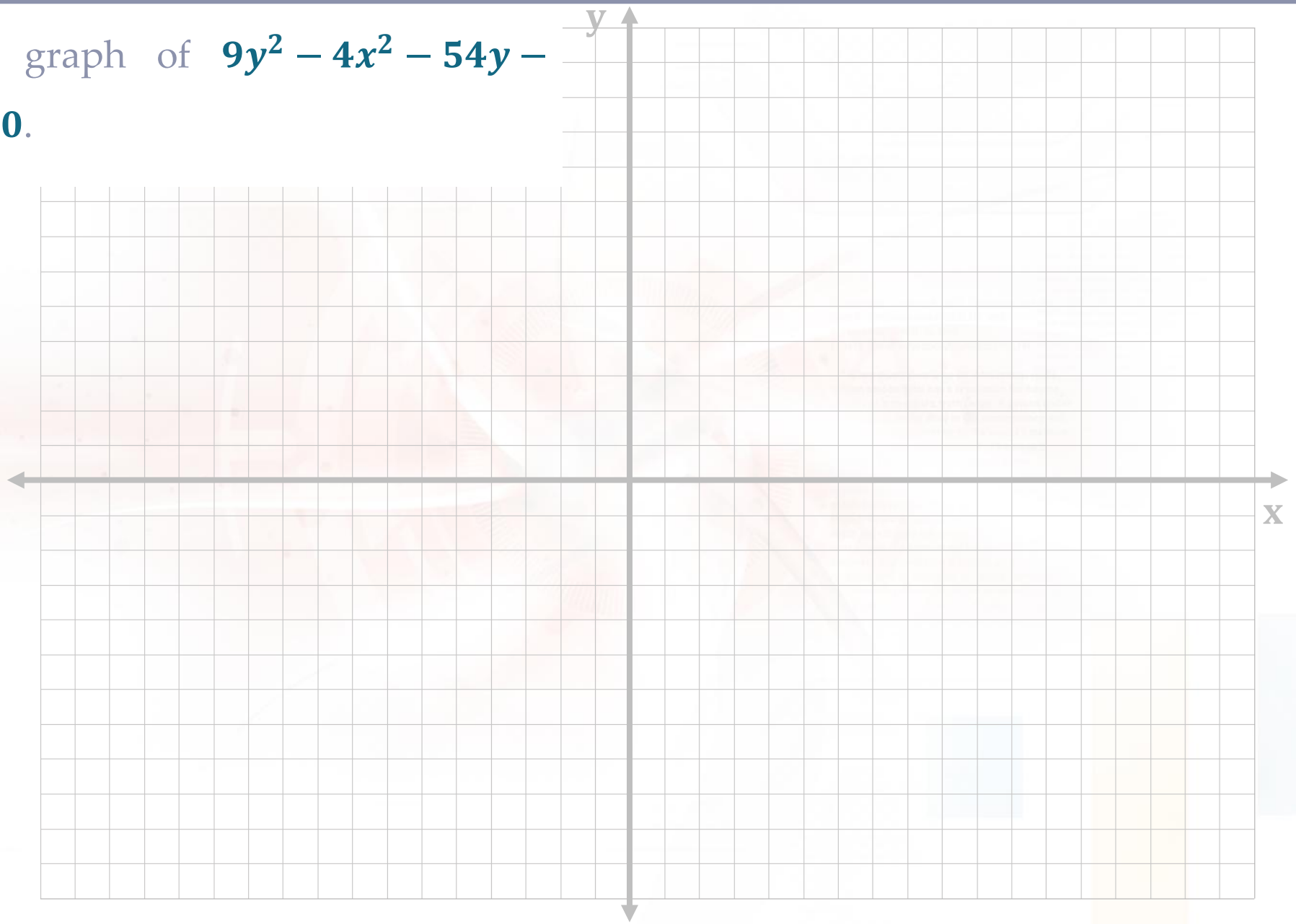

SEATWORK